# Sean Sheng-Tse Ru

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## EDUCATION

**Georgia Institute of Technology** - Atlanta, GA
M.S. in Computer Science (Visual Analytics) May 2025
B.S. in Computer Science (Database/Networking Technologies and Artificial Intelligence) May 2024

## EXPERIENCE

**Data Analyst @ Capital One** Aug. 2025 - Present

- **Sole owner** of two cross-functional analytics initiatives on the Customer Identity team, supporting product managers, fraud analysts, and compliance teams across Credential Management and 2FA programs
- Built an **end-to-end device events monitoring tool** enabling layered analysis of mobile device attributes across **5,000+ attribute combinations**, used to inform policy updates, device blacklisting decisions, and daily anomaly detection for fraud operations
- Leading a platform migration initiative to replace Tableau with a custom **D3.js-hosted solution**, owning technical design, stakeholder alignment, and change management across multiple business teams
- Develop and maintain **SQL pipelines** across large-scale datasets to support fraud detection, customer identity analytics, and cross-team reporting; independently define business logic for new data requests from PMs and BAs

**Researcher @ The Georgia Tech Visualization Lab** Feb. 2024 - April 2026

*beautiVis Dataset - Published, IEEE PacificVis 2026 (co-first author)*

- Designed and executed an end-to-end data annotation pipeline sourcing, filtering, and structuring **52,836 visualizations** from r/dataisbeautiful (2012–2025) using Python, BeautifulSoup, and GPT-4o via OpenAI API
- Engineered multi-task prompt strategy for GPT-4o to simultaneously classify visualization type, chart category, and topical tags at scale; iterated prompt design to resolve systematic misclassification and achieve **Cohen's κ = 0.939** agreement with human coders
- Designed annotation taxonomy mapping **600+ raw chart types** to 12 MASSVIS categories; managed label normalization, post-processing pipelines, and quality validation across 50k+ images for **~$75–100 in API costs**
- Dataset publicly released on Hugging Face and OSF, enabling academic research in information visualization design, saliency modeling, chart classification, and LLM benchmarking

*Charting and Quantifying Visual Complexity:*

- Developed a visualization taxonomy and rubric for labeling **2,000+ entries** in the MASSVIS dataset; presented findings at IEEE VIS 2024 (accepted poster) using R statistical analysis on visual complexity variation across chart types

**Data Analyst Intern @ Capital One** June 2024 - Aug. 2024

- Developed a Tableau dashboard of metrics, graphs, and tables, enabling internal clients to filter and analyze **2+ years of Bankruptcy Complaints data**
- Collaborated with compliance testers, risk management, and business analysts to define output columns and parameters to design and develop SQL and Python scripts for pulling and filtering Complaints and COAF Account Origination data
- Partnered with software engineers on UI enhancements, documenting dependencies and suggesting improvements to streamline the onboarding process of test scripts to the internal testing platform for future DAs

**Data Science Intern @ Optimized Payments** May 2023 - Aug. 2023

- Wrote Python Script to extract data from PDF Files using Tabula
- Used Tableau to visualize and analyze interchange fees and sales data for 4 clients
- Performed data mapping in Snowflake and DBeaver, and developed a Python Script to compare data between SQL and Snowflake to find discrepancies in data migration to Snowflake

## SKILLS

**Languages & Libraries:** Python (Pandas, NumPy, Seaborn, Plotly, PyTorch, scikit-learn, TensorFlow, Tabula), R (ggplot2, Tidyverse), JavaScript (D3.js), HTML/CSS, SQL
**Tools & Platforms:** Tableau, Snowflake, PostgreSQL, MySQL, DBeaver, Git, OpenAI API, BeautifulSoup, Jupyter Notebook

## PUBLICATIONS

Kylie Lin*, **Sean Sheng-Tse Ru***, Minsuk Chang, Cindy Xiong Bearfield. "beautiVis: An Annotated Visualization Dataset from Reddit's r/dataisbeautiful." IEEE PacificVis 2026, Short Paper. (*co-first authors)

Kylie Lin, **Sean Sheng-Tse Ru**, David N. Rapp, Hui Guan, Cindy Xiong Bearfield. "What Makes a Visualization Visually Complex?" *ACM CHI 2025, Extended Abstracts.*